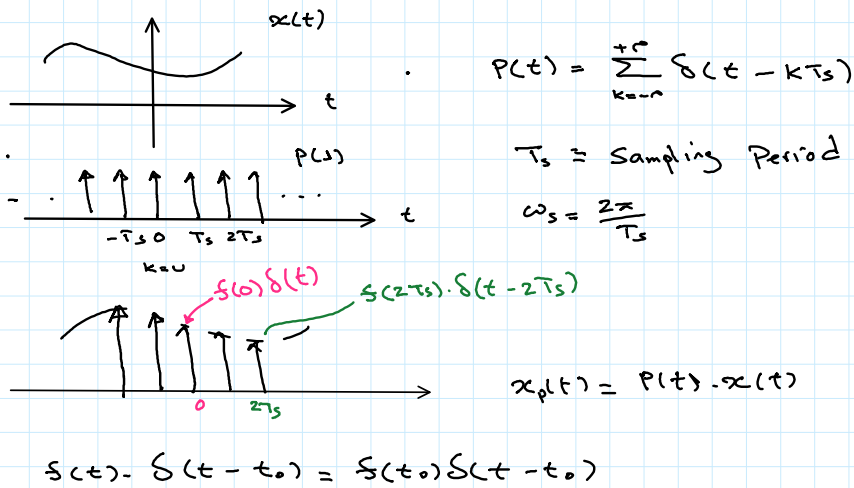
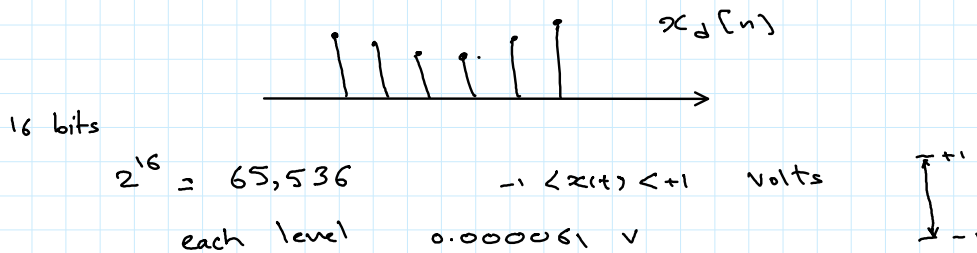


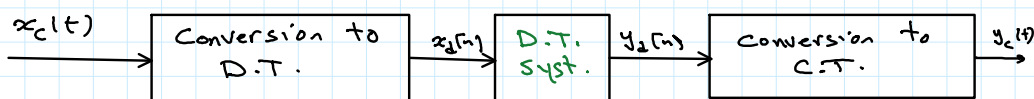
C.T. $x(t) \xleftrightarrow{\text{CTFT}} X(j\omega) \quad -\infty < \omega < +\infty$
 D.T. $x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) \quad \text{periodic, } -\pi < \omega < +\pi$



Quantization



Discrete-Time Processing of C.T. Signals



$$x_c(t) \Big|_{t=nT_s} = x_d[n] = x(t=nT_s) = x(nT_s)$$

$$y_d[n] = y_c(nT_s)$$

ω frequency variable for C.T.

Ω " " " " D.T. $x_d[n]$
 $X_d(e^{j\Omega})$

F.T. $\{x_c(t)\} = X_c(j\omega)$
 $y_c(t) = Y_c(j\omega)$

DTFT $\{x_d[n]\} \equiv X_d(e^{j\Omega})$
 $y_d[n] \quad Y_d(e^{j\Omega})$

$$x_p(t) = x_c(t) \cdot p(t) \longleftrightarrow X_p(j\omega) \quad \text{C.T.}$$

$$\sum_{n=-\infty}^{+\infty} \delta(t - nT_s)$$

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(t) \delta(t - nT_s) = \sum_{n=-\infty}^{+\infty} x_c(nT_s) \delta(t - nT_s)$$

F.T.

$$X_p(j\omega) = \sum_{n=-\infty}^{+\infty} x_c(nT_s) e^{-jn\omega T_s} \quad (I)$$

$x_d[n]$ is a D.T. signal

$$\text{DTFT} \{x_d[n]\} = X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n] e^{-jn\Omega}$$

$$x_d[n] = x_c(nT_s)$$

$$= \sum_{n=-\infty}^{+\infty} x_c(nT_s) e^{-jn\Omega} \quad (II)$$

Compare I & II

$$\Omega = \omega T_s \Rightarrow \omega = \frac{\Omega}{T_s}$$

$$x_p(t) = x_c(t) \cdot \sum_{n=-\infty}^{+\infty} \delta(t - nT_s) \longleftrightarrow X_p(j\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} X(j(\omega - n\omega_s))$$

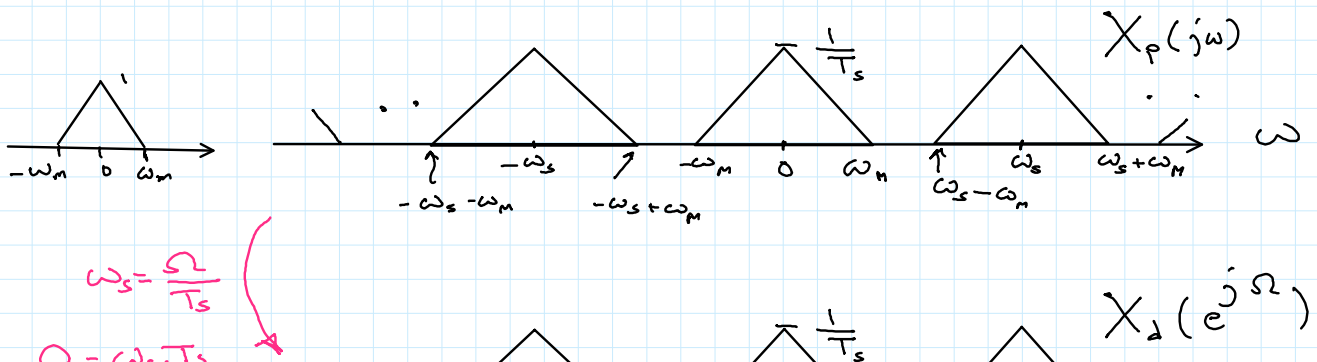
$$x(t) \cdot p(t) \longleftrightarrow \frac{1}{2\pi} \int X(j\omega) \cdot \text{F.T.}\{p(t)\} d\omega$$

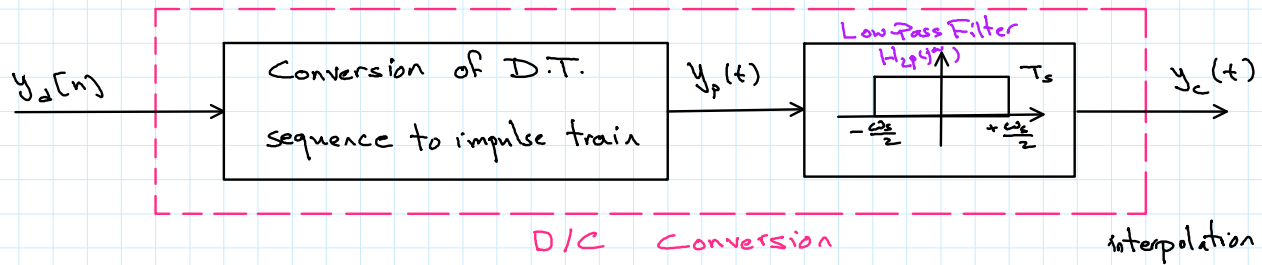
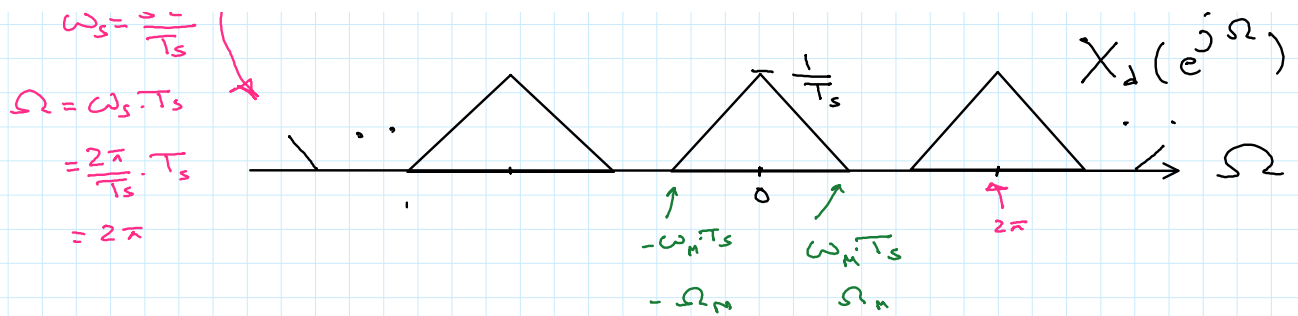
$$p(j\omega) = \sum_{n=-\infty}^{+\infty} \delta(\omega - n\omega_s)$$

$$T_s = \frac{2\pi}{\omega_s}$$

$$\omega_s = \frac{2\pi}{T_s}$$

$$p(j\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} X(\omega - n\omega_s)$$

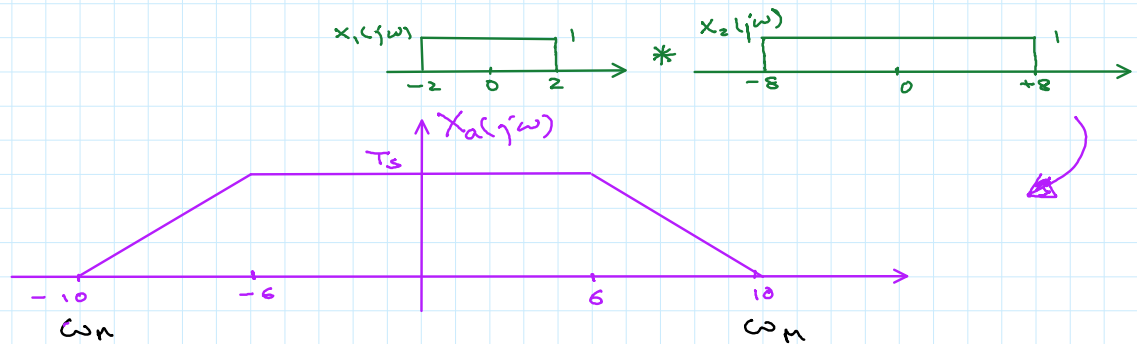




Example:

$$x_a(t) = T_s \left[\underbrace{\frac{\sin(2t)}{\pi t}}_{x_1(t)} * \underbrace{\frac{\sin(8t)}{\pi t}}_{x_2(t)} \right]$$

$$X_a(j\omega) = \frac{1}{2\pi} \{ X_1(j\omega) * X_2(j\omega) \}$$



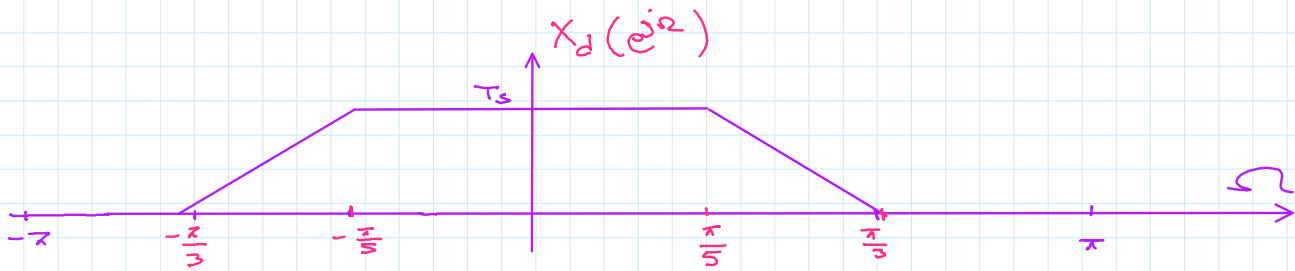
Sample $x_a(t)$ with $T_s = \frac{2\pi}{6} \Rightarrow \omega_s = \frac{2\pi}{T_s} = 60 \text{ rad/s}$

$$x_a(t) \big|_{t=nT_s} = x_a(nT_s) = x_a\left(n\frac{2\pi}{6}\right)$$

Since $\omega_s > 2\omega_m \Rightarrow 60 > 2(10)$
 then **no aliasing**

$\omega_a = 6$ is mapped to $\Omega = \omega_a T_s = (6) \left(\frac{2\pi}{60}\right) = \frac{\pi}{5}$

$\omega_a = 10$ " " " $\Omega = (10) \frac{2\pi}{60} = \frac{\pi}{3}$

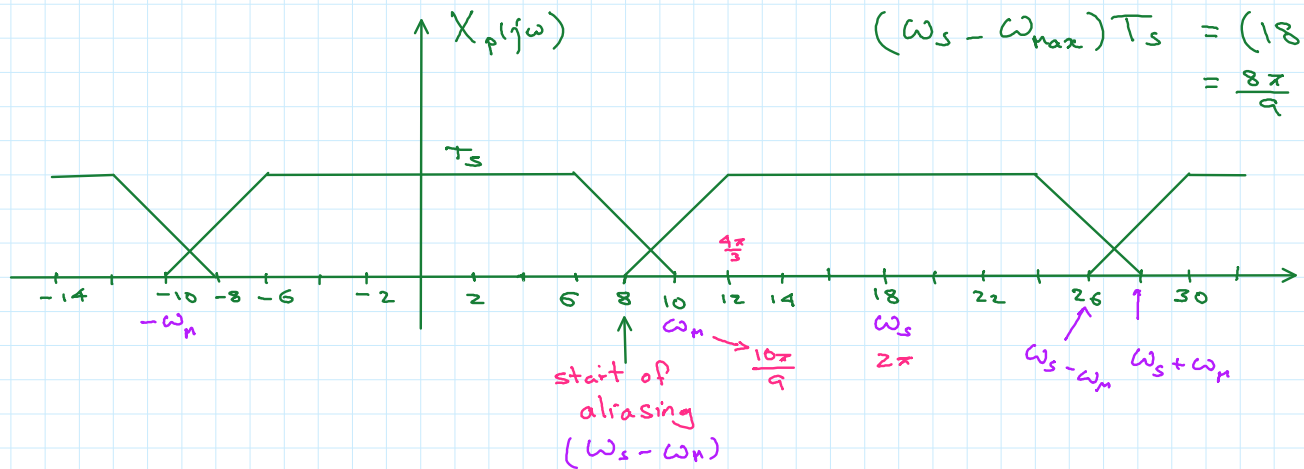


II) if $T_s = \frac{2\pi}{18}$

$$\omega_s = \frac{2\pi}{T_s} = 2\pi \cdot \left(\frac{18}{2\pi}\right) = 18 < 20$$

aliasing occurs (it starts at

$$(\omega_s - \omega_{\max})T_s = (18 - 10) \frac{2\pi}{18} = \frac{8\pi}{9}$$



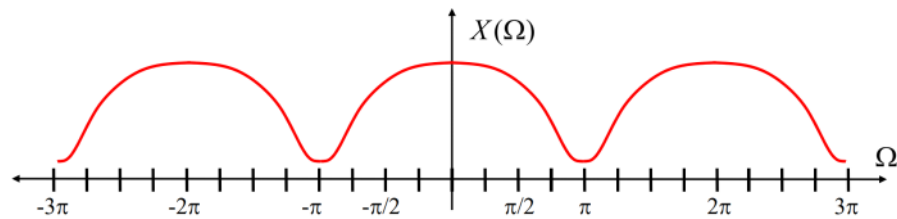
$$\Omega = (\omega_s - \omega_n)T_s = (8) \frac{2\pi}{18} = \frac{8\pi}{9}$$

DTFT & DFT

if $x[n]$ is a finite duration, its DTFT is

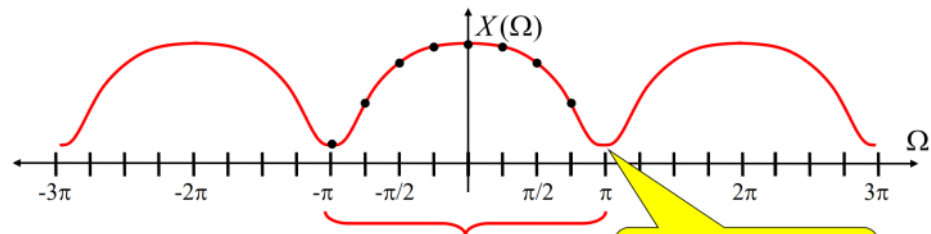
$$X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} = \sum_{n=0}^{N-1} x[n]e^{-j\Omega n}$$

we can leave out terms that are zero



We are only interested in this range...
Everywhere else it just repeats periodically

Now suppose we take the numerical data $x[n]$ for $n = 0, \dots, N-1$
and just compute this DTFT at a finite number of Ω values (8 points here...
but in practice we'd do it MANY more points... thousands of points!)



Region of Interest

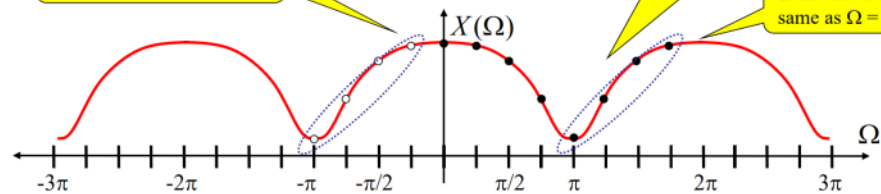
We leave this point out
because it is always the
same value as at $\Omega = -\pi$!!

Now, even though we are interested in the $-\pi$ to π range,
we now play a trick to make the later equations easier...

We don't compute points at
negative Ω values...

But, instead compute their
"mirror images" at Ω values
between π and 2π

Don't need...
same as $\Omega = 0$



So say we want to compute the DTFT at M points, then choose

$$\Omega_k = k \frac{2\pi}{M}, \text{ for } k = 0, 1, 2, \dots, M-1$$

Spacing between computed Ω values

In other words:

$$\Omega_0 = 0, \quad \Omega_1 = \frac{2\pi}{M}, \quad \Omega_2 = 2 \frac{2\pi}{M}, \quad \dots, \quad \Omega_{M-1} = (M-1) \frac{2\pi}{M}$$

So... Given N DFT points $X[k]$ for $k = 0, \dots, N-1$
Compute N signal data points using:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N} \quad n = 0, 1, 2, \dots, N-1$$

**Inverse DFT
(IDFT)**

Compare to the DFT... a remarkably similar structure:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N} \quad k = 0, 1, 2, \dots, N-1$$

DFT

→ 10/13